

CHAPTER 1

LOGIC AND PROOFS

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1.3.1 Predicates

Which statements are propositions:

- Minh loves ice cream.
- X loves ice cream.
- Everyone loves ice cream.
- Someone loves ice cream.

YES

NO

YES

YES

1.3.1 Predicates

Which statements are propositions:

$$3 + 2 = 5$$

YES

$$X + 2 = 5$$

NO

$X + 2 = 5$ for any choice of X in $\{1, 2, 3\}$

YES

$X + 2 = 5$ for some X in $\{1, 2, 3\}$

YES

Proposition, YES or NO?

$$12 > 4$$

YES

$$X > 4$$

NO

$X > 4$ for any choice of X in $\{3, 4, 5\}$

YES

$X > 4$ for some X in $\{1, 2, 3\}$

YES

Predicates.

Definition: A predicate, or propositional function, is a function defined on a set U and returns a proposition as its value.

The set U is called the *universe of discourse*.

- We often denote a predicate by $P(x)$
- Note that $P(x)$ is not a proposition, but $P(a)$ where a is some fixed element of U is a proposition with well determined truth value

Predicates

Let $Q(x,y) = "x > y"$

Which statements are propositions:

$Q(x,y)$

NO

$Q(3,4)$

YES

$Q(x,9)$

NO

$Q(x,y)$ is a predicates in two free variables x and y in $\mathbf{R}$

The universal quantifier $\forall$

Let $P(x)$ be a predicate on some universe of discourse U .

Consider the statement : $\forall x P(x)$

say “ $P(x)$ is true for all x in the universe of discourse.”
“for all x , $P(x)$ ”

The proposition $\forall x P(x)$ is called the *universal quantification* of the predicate $P(x)$.

- $\forall x P(x)$ is TRUE if $P(a)$ is true when we substitute x by any element a in U
- $\forall x P(x)$ is FALSE if there is an element a in U for which $P(a)$ is false.

Examples:

- Let $P(x)$ be the predicate $x + 1 > x$, where the universe of discourse are the real numbers.

Is $\forall x P(x)$ a TRUE proposition ?

- Let $Q(x)$ be the predicate $x < 1$, where the universe of discourse are the real numbers.

Is $\forall x Q(x)$ a TRUE proposition ?

In the special case that the universe of discourse, U , is finite, say $U = \{x_1, x_2, x_3, \dots, x_n\}$. Then

$$\forall x P(x)$$

corresponds to the proposition:

$$P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

The existential quantifier $\exists$

Let $P(x)$ is a predicate on some universe of discourse.

The *existential quantification* of $P(x)$ is the proposition:

$$\exists x P(x)$$

say “for some x , $P(x)$ ”

“There exists an element x in the universe of discourse such that $P(x)$ is true.”

$\exists$ is called the *existential quantifier*

$\exists x P(x)$ is FALSE if $P(x)$ is false for every single x .

$\exists x P(x)$ is TRUE if there is an x for which $P(x)$ is true.

Examples:

- Let $P(x)$ be the predicate $x > 3$, where the universe of discourse are the real numbers.

Is $\exists x P(x)$ a TRUE proposition ?

- Let $Q(x)$ be the predicate $x = x + 1$, where the universe of discourse are the real numbers.

Is $\exists x Q(x)$ a TRUE proposition ?



In the special case that the universe of discourse, U , is finite, say $U = \{x_1, x_2, x_3, \dots, x_n\}$. Then

$$\exists x P(x)$$

corresponds to the proposition:

$$P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$$

Precedence of Quantifiers

- The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus.
- Example: $\forall x P(x) \vee Q(x)$ is the disjunction of $\forall x P(x)$ and $Q(x)$.

Negating quantifier propositions

$$\neg \forall x P(x) \equiv \exists x \neg P(x).$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x).$$

General rule: to negate a quantification,

- move negation ($\neg$) to the right,
- change the quantifier from $\forall$ to $\exists$, and from $\exists$ to $\forall$.

12. Let $Q(x)$ be the statement “ $x + 1 > 2x$.” If the domain consists of all integers, what are these truth values?

- | | | |
|---------------------------------|----------------------------|---------------------------------|
| a) $Q(0)$ | b) $Q(-1)$ | c) $Q(1)$ |
| d) $\exists x Q(x)$ | e) $\forall x Q(x)$ | f) $\exists x \neg Q(x)$ |
| g) $\forall x \neg Q(x)$ | | |

13. Determine the truth value of each of these statements if the domain consists of all integers.

- | | |
|----------------------------------|-----------------------------------|
| a) $\forall n(n + 1 > n)$ | b) $\exists n(2n = 3n)$ |
| c) $\exists n(n = -n)$ | d) $\forall n(3n \leq 4n)$ |

14. Determine the truth value of each of these statements if the domain consists of all real numbers.

- | | |
|-------------------------------------|----------------------------------|
| a) $\exists x(x^3 = -1)$ | b) $\exists x(x^4 < x^2)$ |
| c) $\forall x((-x)^2 = x^2)$ | d) $\forall x(2x > x)$ |

- 34.** Express the negation of these propositions using quantifiers, and then express the negation in English.
- a) Some drivers do not obey the speed limit.
 - b) All Swedish movies are serious.
 - c) No one can keep a secret.
 - d) There is someone in this class who does not have a good attitude.
- 35.** Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.
- a) $\forall x (x^2 \geq x)$
 - b) $\forall x (x > 0 \vee x < 0)$
 - c) $\forall x (x = 1)$

Predicates - multiple quantifiers

Example: $P(x,y) = "x > y"$

■ $\forall x P(x,y)$

NOT a proposition

■ $\forall x \forall y P(x,y)$

FALSE proposition

■ $\forall x \exists y P(x,y)$

TRUE proposition

■ $\forall x P(x, 3)$

FALSE proposition

Predicates - the meaning of multiple quantifiers

- “ $\forall x \forall y P(x,y)$ ” means $P(x,y)$ is true for every possible combination of x and y .
- “ $\exists x \exists y P(x,y)$ ” means $P(x,y)$ is true for some choice of x and y (together).
- “ $\forall x \exists y P(x,y)$ ” means for every x we can find a (possibly different) y so that $P(x,y)$ is true.
- “ $\exists x \forall y P(x,y)$ ” means there is (at least one) particular x for which $P(x,y)$ is always true.

Notice: quantifier order is not interchangeable!

Predicates - the meaning of multiple quantifiers

Example: $N(x,y)$ = “ x is sitting next to y ”

$\forall x \forall y N(x,y)$ - everyone is sitting next to everyone else.

False

$\exists x \exists y N(x,y)$ - there are two people sitting next to each other.

True?

$\forall x \exists y N(x,y)$ - every person is sitting next to somebody.

True?

$\exists x \forall y N(x,y)$ - a particular person is sitting next to everyone else.

False

■ Example:

Express the negation of the statement $\forall x \exists y (xy = 1)$ so that no negation precedes a quantifier.

Solution: By successively applying De Morgan's laws for quantifiers in Table 2 of Section 1.4, we can move the negation in $\neg \forall x \exists y (xy = 1)$ inside all the quantifiers. We find that $\neg \forall x \exists y (xy = 1)$ is equivalent to $\exists x \neg \exists y (xy = 1)$, which is equivalent to $\exists x \forall y \neg (xy = 1)$. Because $\neg (xy = 1)$ can be expressed more simply as $xy \neq 1$, we conclude that our negated statement can be expressed as $\exists x \forall y (xy \neq 1)$. 

TABLE 2 Rules of Inference for Quantified Statements.

<i>Rule of Inference</i>	<i>Name</i>
$\frac{\forall x P(x)}{\therefore P(c)}$	Universal instantiation
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$	Universal generalization
$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$	Existential instantiation
$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$	Existential generalization

- Show that the premises “A student in this class has not read the book,” and “Everyone in this class passed the first exam” imply the conclusion “Someone who passed the first exam has not read the book.”

Summary

TABLE 1 Quantifiers.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

TABLE 2 De Morgan's Laws for Quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

TABLE 1 Quantifications of Two Variables.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$?

Show that $\neg\forall x(P(x) \rightarrow Q(x))$ and $\exists x(P(x) \wedge \neg Q(x))$ are logically equivalent.

Method of proving

- Direct proof
- Proof by contraposition

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

- Proof by contradiction

Prove $p \rightarrow q$:

assume that p and $\neg q$ are true

show a contradiction

- Example 1 : Prove that if n is an integer and $3n + 2$ is odd, then n is odd.
- Example 2: Prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.